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DETAIL SPECIFICATION

ANTIMONY SULFIDE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for all types and classes of antimony sulfide for use in ammunition.

1.2 Classification. Antimony sulfide is of the following types and classes as specified (see 6.2).

1.2.1 Type.

Type I	-	0.2 percent free sulfur, maximum
Type II	-	0.02 percent free sulfur, maximum

1.2.2 Class.

Class 1	-	100 mesh, nominal
Class 2	-	Fine
Class 3	-	100 mesh, nominal, coarse
Class 4	-	Lump
Class 5	-	140 mesh, nominal

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at (https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B215	-	Standard Practices for Sampling Metal Powders
ASTM D1193	-	Standard Specification for Reagent Water
ASTM E11	-	Standard Specifications for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E287	-	Standard Specification for Laboratory Glass Graduated Burets
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Material.

3.2.1 Physical characteristics. Antimony sulfide shall be either natural or synthetic. Antimony sulfide shall have the characteristic shining structure resembling elongated irregular prisms, steel-grey color, and hardness of the naturally occurring crystalline mineral.

3.2.2 Chemical composition. Antimony sulfide shall conform to the requirements shown in Table I.

TABLE I. Chemical composition.

Requirement	Type I	Type II
Antimony (Sb), % minimum (min)	70.5	70.5
Total sulfur (S), % min	24.8	24.8
Insoluble material, % maximum (max)	1.0	1.0
Acidity, % max	0.01	0.01
Lead (Pb), % max	0.15	0.15
Iron (Fe), % max	0.5	0.5
Arsenic (As), % max	0.06	0.06
Free sulfur, % max	0.2	0.02
Moisture, % max	0.20	0.20

3.2.3 Granulation. Antimony sulfide shall comply with the granulation requirements specified in Table II when determined as specified. All percentages shall be by weight using U.S. Alternative Sieves conforming to ASTM E11. The powder shall pass through the required sieves readily without balling or the particles clinging together.

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TABLE II. Granulation requirements.

	Class 1	Class 2	Class 3	Class 4	Class 5
Thru 100	95% min	-	95% min	-	-
Retained on 140	30 to 50%	-	70% min	-	-
Thru 140	-	97% min	-	-	99.0% min
Retained on 200	-	80 to 95%	-	-	95.0% min
Thru 200	20 to 30%	-	20% max	-	-
Note: Granulation requirements have not been established for the Class 4 lump form.					

3.3 Workmanship. Antimony sulfide shall be uniform in quality and free from lumps (except for Class 4), grit, visible impurities, and foreign matter such as dirt, rust, paint, wooden particles or any other foreign objects. Pulverized antimony sulfide shall be free from extraneous material as indicated by visual examination under a magnification between 100x to 1000x that clearly shows the crystal structure. The Class 4 lump form of the antimony sulfide shall be clean and free from adherent gangue material or scale.

4. VERIFICATION

TABLE III. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 3 – Examination 2 – Demonstration 4 – Test		Classes of verification A – First article B – Conformance						
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article			X	X	X		4.2
3.1.2	Conformance			X	X		X	4.3
3.2.1	Physical characteristics			X		X	X	4.4.1
3.2.2	Antimony, % min	X				X	X	4.4.2
3.2.2	Total sulfur, % min	X				X	X	4.4.3
3.2.2	Insoluble material, % max	X				X	X	4.4.4
3.2.2	Acidity, % max	X				X	X	4.4.5
3.2.2	Lead, % max	X				X	X	4.4.6
3.2.2	Iron, % max	X				X	X	4.4.6
3.2.2	Arsenic, % max	X				X	X	4.4.6
3.2.2	Free sulfur, % max	X				X	X	4.4.7
3.2.2	Moisture, % max	X				X	X	4.4.8
3.2.3	Granulation	X			X	X	X	4.4.9
3.3	Workmanship			X		X	X	4.4.10

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4.1 Classification of inspection. The verification requirements specified herein are classified as follows:

a. First article (see paragraph 4.2).
a. Conformance (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table IV. Sampling procedures shall be conducted in accordance with 4.3.2. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

4.2.1 First article rejection. The first article sample shall be rejected if any of the inspections/tests in Table IV fails to meet any of the requirements.

TABLE IV. First article inspection.

Classification	Examination of Test	Requirement Paragraph	Inspection Method Reference
Critical	None defined		
Major			
101	Physical characteristics	3.2.1	4.4.1
102	Antimony	3.2.2	4.4.2
103	Total sulfur	3.2.2	4.4.3
104	Insoluble material	3.2.2	4.4.4
105	Acidity	3.2.2	4.4.5
106	Lead	3.2.2	4.4.6
107	Iron	3.2.2	4.4.6
108	Arsenic	3.2.2	4.4.6
109	Free sulfur	3.2.2	4.4.7
110	Moisture	3.2.2	4.4.8
111	Granulation	3.2.2	4.4.9
Minor			
201	Workmanship	3.3	4.4.10

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph “Formation and identification of lots or batches.” Each lot shall consist of that quantity of antimony sulfide that has been subjected to the same physical processes intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of antimony sulfide produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2 Antimony sulfide sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the antimony sulfide is packaged, or at the receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

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4.3.2.1 Sampling at the supplier/distributor. Samples shall be collected from the beginning, middle and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 500 grams shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected. Before sampling, the contents of each container (selected for sampling) shall be thoroughly blended for 30 minutes by rolling and turning end for end. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

4.3.2.2 Sampling at the receiving facility. Antimony sulfide packaged in shipping containers (e.g., drums, boxes, etc.) shall be sampled using Verification Level II of MIL-STD-1916. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. This material shall be placed in a secondary airtight container and mixed to form a composite sample of not less than 500 grams (g) for each antimony sulfide lot received. When lots comprise three or less containers, all containers in the lot shall be selected. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.3.3. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.

4.3.3 Classification of characteristics. Antimony sulfide shall be subjected to the inspections according to 4.3.3.1.

4.3.3.1 Antimony sulfide.

Category	Examination of Test	Verification Level	Requirement Paragraph	Inspection Method Reference
Critical	None defined			
Major				
101	Physical Characteristics	4.3.2	3.2.1	4.4.1
102	Antimony, % min	4.3.2	3.2.2	4.4.2
103	Total Sulfur, % min	4.3.2	3.2.2	4.4.3
104	Insoluble material, % max	4.3.2	3.2.2	4.4.4
105	Acidity, % max	4.3.2	3.2.2	4.4.5
106	Lead, % max	4.3.2	3.2.2	4.4.6
107	Iron, % max	4.3.2	3.2.2	4.4.6
108	Arsenic, % max	4.3.2	3.2.2	4.4.6
109	Free sulfur, % max	4.3.2	3.2.2	4.4.7
110	Moisture, % max	4.3.2	3.2.2	4.4.8
111	Granulation	4.3.2	3.2.2	4.4.9
Minor				
201	Workmanship	4.3.2	3.3	4.4.10

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4.4 Test methods and procedures. Distilled water conforming to ASTM D1193 and reagent grade materials/chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant.

4.4.1 Physical characteristics. Place a few grams of the sample on a glass slide and examine the material under a 100x to 1000x power microscope such that a representative sample size is visible. Visually examine the sample for conformance.

4.4.2 Determination of antimony.

4.4.2.1 Antimony by titration with potassium permanganate.

4.4.2.1.1 Equipment and supplies.

a. Class A volumetric flasks and pipettes
b. Graduated cylinders
c. Erlenmeyer flasks, suggested size of 250 milliliters (mL)
d. Class A burette (conforming to ASTM E287), 50 mL
e. Analytical balance with 200 grams (g) capacity and 0.1 milligrams (mg) resolution
f. Hot plate, with temperature setting up to at least 300°Celsius (C)
g. Automated titrator with accessories for redox titration (optional)

4.4.2.1.2 Standards and reagents.

4.4.2.1.2.1 Reagents. All materials used shall be reagent grade.

a. Water, deionized (DI) or distilled, conforming to ASTM D1193 Type I or II
b. Sulfuric acid (H_2SO_4), concentrated, 95 to 98% weight (wt)
c. Ammonium sulfate ($(NH)_2SO_4$), 99 %wt
d. Sodium sulfite (Na_2SO_3), 98 %wt
e. Phosphoric acid (H_3PO_4), concentrated, 85 %wt
f. Hydrochloric Acid (HCl), concentrated, 35-38 %wt
g. Hydrochloric Acid (HCl), diluted, 1:1 by mixing equal volumes of concentrated HCl and water
h. Potassium permanganate ($KMnO_4$), 99 %wt, (see 4.4.2.1.3.1)
i. Potassium permanganate solution, 0.1 molar (M), commercial, standardized (see 4.4.2.1.3.1)
j. Potassium permanganate solution, 0.02 M, standardized against antimony, see below (see 4.4.2.1.3.3)
k. Antimony (Sb), powder, 99.9+ %wt, trace metals basis, certified material
l. Antimony (III) sulfide (Sb_2S_3), powder, 99.9+ %wt metals basis, reference material

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4.4.2.1.3 Solutions preparation and standardization.

4.4.2.1.3.1 Potassium permanganate solutions preparation. The potassium permanganate solution may be prepared from solid potassium permanganate or diluted from commercially available 0.1 M solution with certified concentration.

4.4.2.1.3.2 $KMnO_4$ 0.1 M stock solution, preparation from solid $KMnO_4$. Dissolve 15.9 g of potassium permanganate in 900 mL water. Heat the solution at 80 to 90ºC for 1 hour, allow to cool and filter through a sintered-glass or glass-fiber filter. Make up to 1 liter (L) with water and mix well. Store the solution in a clean, glass-stoppered bottle, protected from light.

4.4.2.1.3.3 $KMnO_4$, 0.02 M solution. Pipet 100 mL of the $KMnO_4$ 0.1 M stock into a 500-mL volumetric flask. Fill to the mark with distilled water and mix well. Standardize this solution against high-purity antimony metal before use as described below (see 4.4.2.1.3.5). Alternately, standardize against sodium oxalate using commonly accepted procedure, or purchase commercially available standardized 0.02M (0.1N) KMnO4.

4.4.2.1.3.4 Expiration. The maximum recommended shelf life of stock solutions is one (1) year from date of initial preparation when stored at 5 to 25ºC, tightly sealed and protected from light. Stock solution shall be replaced after the expiration date or in accordance with manufacturer recommendations.

4.4.2.1.3.5 Potassium permanganate titrant standardization. Three (3) replicates shall be prepared from high-purity antimony metal as follows. Weigh 200.0 ± 5.0 mg of antimony powder into a 250-mL conical flask and proceed with sample digestion and titration in the manner described below in the sample preparation and analysis section (see 4.4.2.1.5).

Calculate the titer value of 0.02 M $KMnO_4$:

$$T = \frac{M_{Sb} * Purity}{(V_{Sb} - V_b) * 100}$$

Where:

Purity = purity grade of antimony powder from commercial Certificate of Analysis (CoA), for example 99.99 %wt
V_{Sb} = volume of titrant used for antimony, mL
V_b = volume of titrant used for blank, mL
M_{Sb} = antimony sample mass, mg

Determine the average and relative standard deviation (RSD) of the titer value over the three replicates. The RSD shall be less than 1%.

4.4.2.1.4 Preparation of samples, reference samples, and reagent blank. Prior to analysis a representative 10 g portion of the sample shall be dried at 105ºC for 2 hours, (e.g. material used for moisture determination in 4.4.8).

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Two (2) 300 mg replicates of each sample shall be prepared and analyzed. Method accuracy shall be verified with two (2) 300 mg replicates of high-purity reference antimony sulfide. A reagent blank shall be carried through the entire procedure.

4.4.2.1.5 Sample preparation and analysis. Weigh 300.0 ± 5.0 mg of sample into a 250-mL Erlenmeyer flask (see 4.4.2.1.1.c).

Add 3 g of ammonium sulfate and 15 mL sulfuric acid to the flask, and heat the mixture on the hot plate at ~290°C until the sample is completely dissolved, the solution is clear and colorless with at most some dark orange droplets of sulfur. Heat solution to the formation of white fumes within the flask and fume for 5 minutes. Let the solution cool enough to handle. In small increments add 30 mL of water (to avoid splashing from the heat generated). Add 0.5 g of sodium sulfite, mix well, bring the solution to a boil, and continue to boil for a few minutes. Immediately add 5 mL of phosphoric acid and reheat the solution for a few minutes. Add 20 mL of HCl (1:1), mix well until white precipitate is fully dissolved (heat the solution if necessary). Make up the volume to 100 mL with water. Cool the warm solution in an ice-water bath for at least 20 minutes.

Titrate the cold sample solution with 0.02 M solution of potassium permanganate. Add the first ~1 mL of the titrant dropwise, mixing well after each addition. Proceed with faster titrant addition and slow down toward the end of titration, when pink color starts to spread over the entire solution. Mix the sample solution well during titration. End the titration when the sample solution turns pale pink that persists for 10 seconds. Record the end volume of the titrant.

Titrate the reagent blank in the same manner until the solution turns pinkish-brown.

Alternately, an automated titrator with accessories for redox titration and potentiometric endpoint detection can be used to analyze all samples. Instrument parameters should be optimized prior to running analyses.

4.4.2.1.6 Reference sample preparation. Weigh 300.0 ± 5.0 mg of antimony sulfide high-purity reference material into a 250-mL Erlenmeyer flask (see 4.4.2.1.1.c). Proceed in the manner described in the sample preparation and analysis section (see 4.4.2.1.5).

4.4.2.1.7 Sample analysis and results calculation. Calculate the weight percent of antimony in samples and reference samples as follows:

$$\%\text{wt Antimony} = \frac{(V\text{-}V_b)\times T}{M}\times 100$$

Where:

V = volume of titrant used for sample, mL
V_b = volume of titrant used for blank, mL
M = sample mass, mg
T = average titer of 0.02 M $KMnO_4$, mg/mL

The calculated weight percent of antimony in reference sample shall be compared with the expected value:

$$\text{Expected \%wt antimony} = 0.7168\times \text{CoA \%wt}$$

Where:

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0.7168 = coefficient (weight fraction of Sb in Sb_2S_3)
CoA %wt = purity grade of antimony sulfide from commercial Certificate of Analysis, e.g. 99.9 %wt

For the results to be considered valid, the absolute difference (in units of %wt antimony) between the replicates shall be no more than 0.50. The absolute difference between the averaged reference sample result and its expected value shall be no more than 0.60 %wt.

Calculate and report the average of the two sample replicates.

4.4.2.2 Determination of antimony (alternate method). Dry a portion of the material (ground to pass a 60 mesh sieve, if necessary) by heating at 105ºC for 1 hour. Cool in a desiccator for 30 minutes. Weigh a 0.2 g sample to 0.1 of a mg and transfer to a wide-mouthed 300-mL flask containing 3 g of ammonium sulfate and 20 mL of sulfuric acid. Heat strongly over a free flame until the sample is completely decomposed and any sulfur which is present is volatilized. Continue heating for an additional 10 to 15 minutes. Cool to room temperature. Add 60 mg of flowers of sulfur to the sample and blank, and heat on a hot plate (surface temperature approximately 360ºC) to the formation of white fumes within the flask. Continue heating for an additional 5 minutes. Transfer the flask to a position over a free flame and heat strongly for 1 minute, during which time the sulfur is volatilized from the sides and neck of the flask by using an auxiliary burner. A small sulfur bead should remain in the solution. Remove the flask to an asbestos board, wait 10 seconds (but not more than 20 seconds), and add directly to the flask 20 mL of a mixture made of equal parts of sulfuric acid and 85% phosphoric acid. Mix well and allow to cool to room temperature. Wash down cautiously with 40 mL of water, heat to boiling, and boil gently for 5 minutes. Cool in an ice bath to below 10ºC. Wash down with 50 mL of cold dilute HCl (1:1), shake until the white precipitate is completely dissolved, and dilute to 300 mL with ice water. The temperature of the solution should be about 10ºC. If necessary, cool to this temperature in an ice bath. Titrate with 0.05 normal (N) potassium permanganate solution to the first definite pink color that persists for 15 seconds. Deduct a blank carried through all steps of the procedure. Calculate as follows:

$$\text{Percent antimony} = \frac{6.088 \times V \times N}{W}$$

Where:
V = volume of potassium permanganate solution corrected for the blank, mL
N = normality of potassium permanganate solution
W = weight of sample, g

4.4.3 Determination of total sulfur. Weigh a 0.5000 ± 0.0100 g sample to 0.1 mg and transfer to a 250-mL beaker or flask. Add a mixture of 1 mL of deionized water, 9 mL of glacial acetic acid, and 6 mL of bromine. Cover with a watchglass and allow to stand at room temperature for at least 16 hours. Add 20 mL of HCl, warm gently for 1 hour and then evaporate almost to dryness. Add 20 mL of a 20% HCl solution (3:2 concentrated HCl:H_2O volume/volume (v/v)) and 2.0 g of tartaric acid and dilute to approximately 150 mL by the addition of water at a temperature of 80 to 90ºC. Precipitate the antimony by the addition to the

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warm solution, over a period of 20 minutes, of 0.5 g of powdered aluminum, a little at a time. Filter through a medium filter (e.g. Whatman #40) and wash the beaker, filter and residue with five 30 mL portions of hot water. Dilute the combined filtrate and washings to approximately 500 mL with water, cover and heat to boiling. Remove from heat, and to the hot solution add dropwise 10 mL of a 10% solution of barium chloride. Cover and boil for 15 minutes, digest on a steam bath for 2 hours, and allow to stand 4 hours or overnight. Vacuum-filter through a fine-porosity porcelain filtering crucible pre-weighed to 0.1 mg and wash the beaker and crucible with five (5) 30 mL portions of hot water. Dislodge any precipitate from the beaker walls with a ~3x3 square centimeters (cm^2) piece of ashless filter paper. Transfer paper and residue to the crucible with a stream of water. Ignite crucible gently at first and then at a cherry-red heat (alternately, heat in a furnace set to 900ºC for 2 hours), cool on a heat-proof surface to ~100ºC and then to room temperature in a desiccator, and re-weigh to 0.1 mg. Calculate as follows:

$$\text{Percent total sulfur} = \frac{13.74 \times (R-C)}{W}$$

Where:

R = weight of crucible and $BaSO_4$ residue after ignition, g

C = tare weight of crucible, g

W = sample weight, g

4.4.4 Determination of insoluble material. Weigh a 5 g sample into a 400-mL beaker and add 150 mL of concentrated hydrochloric acid. Digest on a steam bath until no reaction is evident and until most of the hydrogen sulfide has been expelled. Very slowly add 50 mL of concentrated nitric acid and allow to stand on a hot plate (surface temperature 160-200ºC) for 30 minutes to effect solution of the antimony sulfide and expulsion of hydrogen sulfide. Filter through a tared porcelain filtering crucible of fine porosity and wash with three (3) 25 mL portions of hot water. Dry the crucible at 105ºC for 30 minutes and heat by means of a gas flame for 5 minutes in order to remove any sulfur present. Cool in a desiccator and weigh. Calculate as follows:

$$\text{Percent insoluble material} = \frac{\text{g of residue}}{\text{g of sample}} \times 100$$

4.4.5 Determination of acidity. If necessary, grind a portion of the sample so that it passes through a No. 100 U.S. Alternate Sieve and mix thoroughly. Weigh a 10 g sample into a 250-mL beaker and add 100 mL of hot water. Heat to boiling while stirring, and filter immediately through a fine porosity filter, e.g. Whatman #42 or finer (suspended sample fines obscure indicator color). Add 2 drops of 0.1% weight/volume (w/v) methyl orange indicator, and if the solution is acid (pink), titrate with 0.01 M potassium hydroxide (KOH) till an orange color persists for at least 30 seconds. Calculate acidity as follows:

$$\text{Percent sulfuric acid} = \frac{4.90 \times V \times M}{W}$$

Where:

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V = volume of potassium hydroxide solution, mL
M = molarity of potassium hydroxide solution
W = weight of sample, g

Prepare and standardize 0.01 M KOH as follows. Dissolve 0.66 ± 0.02 g of reagent grade KOH in one liter of deionized water, mix well, and fill a 50 mL Class A burette with the solution. In a 125-mL conical flask dissolve 0.0620 ± 0.0050 g reagent grade potassium acid phthalate (KHP) in 50 mL deionized water, then add two drops of 1% w/v phenolphthalein indicator and titrate with the KOH solution to the first appearance of faint pink that lasts at least 30 seconds. Titrate a blank consisting of 50 mL deionized water the same way. Calculate effective molarity, M of KOH as follows:

$$M = \frac{4.90 \times K}{(V\text{-}B)}$$

Where:

K = weight of KHP used, g
V = volume of KOH solution used for KHP, mL
B = volume of KOH solution used for blank, mL

4.4.6 Determination of arsenic, iron and lead by Inductively Coupled Plasma – Optical Emission Spectroscopy (ICP-OES).

4.4.6.1 Equipment and supplies.

a. Inductively coupled, argon-plasma, optical emission spectrometer, dual view
b. Compressed argon, as per ICP-OES manufacturer recommendations
c. Class A volumetric flasks and pipettes
d. Adjustable volume pipettes calibration verified at most a month prior: 0.2 to 1 mL and 1 to 5 mL
e. Analytical balance with 200 g capacity and 0.1 mg resolution.

4.4.6.2 Standards and reagents.

4.4.6.2.1 Reagents.

All materials used shall be reagent or ACS grade, or as specified below.

a. Water, deionized (DI) or distilled, conforming to ASTM D1193 Type I or II
b. Nitric Acid (HNO_3), concentrated, 67-70% (wt), trace-metal grade
c. Hydrochloric Acid (HCl), concentrated, 34-37% wt, trace-metal grade
d. Hydrochloric Acid (HCl) 4% v/v: made by diluting 40 mL of concentrated HCl to 1 L with water.
e. Hydrogen peroxide, 30% weight/weight (w/w)
f. Bromine

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g. Tartaric acid
h. Tartaric acid 20% w/v aqueous solution: made by dissolving 100 ± 1 g of tartaric acid in enough water to make 500 mL of solution

4.4.6.2.2 Standard stock solutions. It is recommended to use commercially available ICP standards as stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after the expiration date or in accordance with manufacturer recommendations. Common, commercially available stock solutions are as follows:

a. Certified As (V) stock solution, 1000 ± 2 mg/L in 1-5% (v/v) HNO_3
b. Certified Fe stock solution, 1000 ± 2 mg/L in 1-5% (v/v) HCl or HNO_3
c. Certified Pb stock solution, 1000 ± 2 mg/L in 1-5% (v/v) HNO_3 or HCl
d. Certified Sb stock solution, 10000 ± 20 mg/L in 1-20% (v/v) HCl or HNO_3, stabilized with tartaric acid
e. Certified S stock solution, 1000 ± 2 mg/L in water (where sulfur is in the form of sulfate)
f. Certified Y stock solution, 1000 ± 2 mg/L in 1-5% (v/v) HNO_3. Yttrium serves as internal standard (IS) during analysis.

Alternately, stock solutions may be prepared in-house from high purity metals or their salts as follows; these materials shall be 99.9% purity or better.

Arsenic (V), 1000 mg/L: dissolve 1.320 ± 0.003 g of arsenic oxide (As_2O_3) in 30 mL of HNO_3, then heat to boiling. Cool to room temperature, transfer quantitatively to a 1 L volumetric flask, and dilute to mark with water.

Iron, 1000 mg/L: dissolve 1.000 ± 0.003 g of elemental iron in 75 mL of HCl by warming on a hot plate. Add 3 mL of 30% hydrogen peroxide; bring to a boil to destroy the peroxide. Cool, then transfer quantitatively to 1 L volumetric flask and dilute to mark with water.

Lead, 1000 mg/L: dissolve 1.000 ± 0.003 g of elemental lead in a mixture of 10 mL HCl and 20 mL nitric acid. Heat and boil to get rid of nitrogen oxides. Cool, then transfer quantitatively to 1 L volumetric flask and dilute to the mark with water.

Antimony, 10000 mg/L: dissolve 10.00 ± 0.01 g of elemental antimony in a mixture of 200 mL of HCl and 50 mL nitric acid. Heat and boil to get rid of nitrogen oxides. Cool, add 10 g of tartaric acid, dissolve, and transfer quantitatively to 1 L volumetric flask and dilute to the mark with water.

Sulfur, 1000 mg/L: In a 1 L volumetric flask, dissolve 4.43 ± 0.01 g of sodium sulfate (Na_2SO_4) in water. Dilute to the mark with water.

Yttrium, 1000 mg/L: dissolve 1.000 ± 0.005 g of elemental yttrium in a mixture of 10 mL HCl and 50 mL of HNO_3. Heat and boil to get rid of nitrogen oxides. Cool, transfer to 1 L volumetric flask and dilute to the mark with deionized water.

The maximum recommended shelf life of stock solutions is one (1) year from date of initial preparation when kept tightly sealed at 5 to 25°C.

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4.4.6.2.3 Preparation of intermediate standard solutions and blanks. The intermediate standards shall be prepared daily as needed.

Arsenic (V), 100 mg/L: pipet 5.00 mL of the As (V) 1000 mg/L stock into a 50 mL volumetric flask. Fill to the mark with 4% v/v HCl and mix well.

Lead, 100 mg/L: pipet 5.00 mL of the lead 1000 mg/L stock into a 50 mL volumetric flask. Fill to the mark with 4% v/v HCl and mix well.

Yttrium, 100 mg/L: dilute 10.0 mL of the 1000 mg/L yttrium stock to 100 mL with 4% v/v HCl.

Calibration blank: place 20 mL of deionized water into 100 mL volumetric flask, add 1.00 mL of yttrium 100 mg/L stock, fill to the mark with 4% v/v HCl and mix well.

4.4.6.3 Calibration and analysis. Prepare the working calibration solutions with concentrations given in Table V by pipetting the indicated volumes of stock or intermediate stock solutions into 100-mL flasks as per Table VI and dilute to the mark with water.

TABLE V. Concentrations of calibration solutions, mg/L.

Element	Sb*	S*	As	Fe	Pb	Y
Calibration blank	0.0	0.0	0.000	0.000	0.000	1.00
Calibration standard 1	700	250	0.300	3.50	0.500	1.00
Calibration standard 2	700	250	0.500	4.50	1.00	1.00
Calibration standard 3	700	250	0.700	5.50	2.00	1.00
Calibration standard 4	700	250	0.900	6.50	3.00	1.00

*Note: Sb and S concentrations given for reference only; these elements are not quantified by the method.

TABLE VI. Volumes (mL) indicated for preparation of calibration solutions.

Element	Sb	S	As	Fe	Pb	Y
Stock concentration, mg/L**	10000	1000	100	1000	100	100
Calibration blank	0	0	0	0	0	1.00
Calibration standard 1	7.0	25.0	0.300	0.350	0.500	1.00
Calibration standard 2	7.0	25.0	0.500	0.450	1.00	1.00
Calibration standard 3	7.0	25.0	0.700	0.550	2.00	1.00
Calibration standard 4	7.0	25.0	0.900	0.650	3.00	1.00

**Note: Actual stock concentrations of commercially obtained solutions or of those prepared in-house shall be used. Concentrations of calibrations standards in Table V shall be adjusted accordingly.

4.4.6.4 Preparation of samples, spiked verification samples, and reagent blank. Two (2) 100 mg portions of each sample shall be prepared and analyzed. Two (2) additional 100 mg aliquots shall be spiked with Pb, Fe, and As, and used to verify method performance. A reagent blank, i.e. 10 mL HNO_3, 0.5 mL bromine, and 20 mL HCl, shall be carried through the entire procedure and its reading shall be subtracted from all samples.

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4.4.6.4.1 Sample preparation. Weigh 100.0 ± 5.0 mg of sample into a 250-mL conical flask. Add 10 mL HNO_3, 0.5 mL bromine, and 20 mL HCl to the flask, and cover with a watch glass. Allow the sample to dissolve completely without heating and with only occasional swirling (approximately 10 minutes). Rinse the cover into the flask with water. Heat the uncovered flask to boiling to remove the bromine (until solution is no longer brown). Cool the solution, add 10 mL of the 20% tartaric acid, mix, and transfer quantitatively to a 100-mL volumetric flask. Add 1.00 mL of Yttrium 100 mg/L stock, dilute to mark and mix well.

4.4.6.4.2 Spiked sample preparation. Weigh 100.0 ± 5.0 mg of sample into a 250-mL conical flask. Prior to adding reagents, add stock solutions of Pb, Fe, and As, as per Table VII. Continue with sample preparation by adding HNO_3, bromine, and HCl and proceeding with the rest of the preparation steps (see 4.4.5.4.1).

TABLE VII. Preparation of spikes.

Element	Pb	Fe	As
Stock concentration, mg/L	100	1000	100
Spike volume, mL	1.00	0.250	0.300
Spiked (added) concentration, mg/L	1.00	2.500	0.300

4.4.6.5 Instrument calibration. Prepare the instrument according to manufacturer instructions. Allow for adequate flush time of the system between successive sample solutions to avoid carryover. Calibrate the instrument using the calibration standards prepared as per Table V. The wavelengths used for each element shall be as per Table VIII, using yttrium as internal standard. The calibrations shall be linear with R^2>=0.999. Analyze the samples in the same run. Run calibration standard #3 to verify instrument stability immediately after calibration and after all samples have been analyzed. If the reported concentrations for this verification standard are significantly different from their expected values (Table V) and cause the samples to fail that would otherwise pass (or vice versa), a correction factor shall be applied to samples and spikes. The correction factor shall equal the quotient of expected concentration and the average reported concentration of the verification standard for the given element.

TABLE VIII. Suggested wavelengths and view modes for ICP-OES element analysis.***

Element	Wavelength, nm	Mode	Referencing elements
As	193	axial	-
Fe	259	radial	-
Pb	220	axial	-
Y (internal standard)	224	axial	As, Pb
Y (internal standard)	371	radial	Fe
***Note: The internal standard wavelengths and view modes are provided as reference per manufacturer recommendations for a ThermoFisher iCAP6500 ICP system. Refer to manufacturer recommendations for the specific equipment being used and adjust for analysis as necessary.			

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4.4.6.6 Sample analysis and results calculation. Aspirate each sample for the time required by the instrument to achieve a stable response. Measure the response and calculate the concentration of each element from its corresponding calibration line, recording the results in mg/L. Calculate the weight percent of each element as follows:

$$\%wt = \frac{(S-B)}{M} \times 10$$

Where:

S = Sample concentration, mg/L
B = blank concentration, mg/L
M = Sample mass, mg
10 is the product of flask volume, i.e. 0.1 L and 100%

The results of sample replicates and spiked sample replicates shall differ by no more than 0.10% wt for Fe, and 0.01% wt for As and Pb. Spike recovery shall be within 10% relative deviation of expected values for all elements (See Table VII). For each reported element, calculate and report the average of the two sample replicates.

4.4.6.7 Determination of iron (alternate).

4.4.6.7.1 Reagents.

a. Water, reagent grade conforming to any type per ASTM D1193; all references to water below shall be interpreted as such.

b. Hydrochloric Acid (HCl), reagent grade 34 to 37 %wt

c. Iron, elemental (wire or filings), 99.99+ %wt metals basis

d. Hydrogen peroxide, reagent grade, 30 %wt

e. Hydroxylamine hydrochloride, reagent grade

f. 1, 10-phenanthroline (ortho-phenanthroline), anhydrous, reagent grade

g. Tartaric acid, 99+ %wt

h. Sodium acetate trihydrate, reagent grade

i. Standard iron solution (1 mL = 0.1 mg iron). Dissolve 1.000 g of iron in 75 mL of hydrochloric acid by warming on a hot plate. Add 3 mL of 30 %wt hydrogen peroxide and boil for 10 minutes to destroy the peroxide. Cool and dilute to 1 L in a volumetric flask. Pipet 50 mL of this solution into a 500-mL volumetric flask and dilute to the mark with water.

j. Hydroxylamine hydrochloride solution (5%). Dissolve 25 g of hydroxylamine hydrochloride in water and dilute to 500 mL.

k. Ortho-phenanthroline solution (0.2%). Dissolve 1.0 g of ortho-phenanthroline in 500 mL of water. Some of the solute may not dissolve; filter or decant prior to use.

l. Tartaric acid solution (10%). Dissolve 50 g of tartaric acid in water and dilute to 500 mL.

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m. Sodium acetate solution (50%). Dissolve 250 g of sodium acetate trihydrate in water and dilute to 500 mL.

4.4.6.7.2 Preparation of calibration curve. Pipette 1, 2, 3, 5, 6, and 7 mL aliquots of standard iron solution into 200-mL volumetric flasks and add 40 mL of water, 4 mL of HCl, and 4 mL of tartaric acid solution. Carry along a reagent blank. Add 10 mL of hydroxylamine hydrochloride solution, mix, and allow to stand for 15 minutes. Add 10 mL of ortho-phenanthroline solution and 20 mL of sodium acetate solution, and dilute to the mark with water. Mix, and allow to stand for 30 minutes. Read the absorbance at 500 nanometers (nm) using a spectrophotometer or colorimeter with a 1 centimeter (cm) cell. Use the reagent blank as instrument baseline (i.e., to establish zero absorbance). Plot milligrams of iron against absorbance; calibration shall be linear and forced through zero with $R^2 > 0.999$.

4.4.6.7.3 Procedure. Transfer a 0.500 ± 0.005 g sample to a 400-mL beaker and add 20 mL of HCl. Cover and boil on a hot plate to dissolve the sample and drive off the hydrogen sulfide. Cool to room temperature, rinse the cover glass into the beaker, and add 20 mL of tartaric acid solution. Quantitatively transfer to a 100-mL volumetric flask. Rinse the beaker three times with 15 mL water and add the rinsate to the flask. Dilute to the mark with water and mix well. NOTE: If there is undissolved material, filter through a medium porosity filter (e.g. Whatman #40); discard the first 10 to 20 mL of the filtrate and collect the next 25 to 70 mL. Pipet a 20 mL aliquot of the clear filtrate into a 200-mL volumetric flask, add 30 mL of water and 10 mL of hydroxylamine hydrochloride solution. Mix and allow to stand 15 minutes, then add 10 mL of ortho-phenanthroline solution and 20 mL of sodium acetate solution. Dilute to the mark with water. Mix, allow to stand for 30 minutes, and then read the absorbance at 500 nm as done previously with standard solutions (the spectrophotometer shall have been blanked with reagent blank beforehand). Convert the readings to milligrams of iron by referring to the calibration curve. Calculate iron content as follows:

$$\text{Percent iron} = \frac{\text{mg iron as read from curve}}{2 \times W}$$

Where:

W = sample weight of antimony sulfide, g

4.4.7 Determination of free sulfur. Weigh a 10 g sample into an extraction thimble approximately 22 by 80 millimeters (mm). Add 75 mL of methylene chloride into a tared 125-mL extraction flask and extract for 4 hours in a Soxhlet apparatus. Disconnect the flask and evaporate the solution to a volume 3 to 5 mL by boiling gently under a hood. Evaporate the residual solvent by aeration at room temperature under a hood. Transfer the flask to a vacuum desiccator or a vacuum oven in which a vacuum of 25 to 30 inches of mercury is maintained for 30 minutes. Weigh and calculate as follows:

$$\text{Percent free sulfur} = \frac{\text{g of residue}}{\text{g of sample}} \times 100$$

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4.4.7.1 Determination of free sulfur (Alternate method). Weigh a 10.0 ± 0.1 g sample into a fine porosity crucible. Record the gross weight to 0.0001 g. Place the crucible in a vacuum filtration adapter and add 15 mL of near-boiling (~45ºC) carbon disulfide (CS_2). Cover crucible and allow five minutes of contact time before applying vacuum. Turn off vacuum, add another 15 mL of hot CS_2, wait five minutes and apply vacuum. Finally, rinse with acetone and suction dry. Place crucible in 60 to 70ºC oven for one hour, cool in a desiccator, and reweigh. Calculate free sulfur as follows:

$$\text{Percent free sulfur} = \frac{100}{W}(\text{S-R})$$

Where:

S = gross weight of crucible and sample at start, g
R = gross weight of crucible and sample at end, g
W = weight of sample

4.4.8 Determination of moisture. Weigh 10 g ± 0.1 g of sample into a tared large weighing dish and record the gross weight to the nearest 0.001 g. Heat at 105ºC for 2 hours, cool in a desiccator and re-weigh. Calculate as follows:

$$\text{Percent moisture} = \frac{\text{g loss in weight}}{\text{g of sample}} \times 100$$

4.4.9 Determination of granulation. Use 8-inch diameter U.S. Alternate Sieves conforming to ASTM E11. For the given class of material, nest the sieve or sieves specified in Table II in order of increasing sieve number (increasing fineness), from top to bottom, and attach a bottom pan.

*Note: If the material class is unknown, all sieves specified in Table II shall be used.

A weighed portion of 100.0 ± 1.0 g of the sample shall be placed on the upper sieve, a lid attached, and the sieve stack shaken for 10 minutes by hand or for 5 minutes by means of a mechanical shaker geared to produce 300 plus or minus 15 gyrations and 150 plus or minus 10 taps of the striker per minute.

The amounts retained on the sieves and the bottom pan shall be weighed and the corresponding percentages of material retained on “On Sieve No.” or passing through “Thru Sieve No.” calculated as follows to determine compliance with section 3.2.3.

$$\text{Retained On Sieve No.} = \frac{100R}{W}\%$$

$$\text{Thru Sieve No.} = \frac{100\text{Sum}(R_B)}{W}\%$$

Where:

R = weight of material retained on specified sieve, g
$\text{Sum}(R_B)$ = total weight of material retained on sieves below specified sieve and

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on bottom pan, g
W = sample weight, g

4.4.10 Workmanship. Take approximately 100 to 200 g of the material and spread the compositions out on a clean white surface. The sample shall be examined by personnel with normal vision or normal corrected vision. The sample shall be uniform in quality and free from visible impurities, foreign matter or other defects. Failure to meet the workmanship inspection or during following production processes shall result in batch rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I is intended for use in the usual priming compositions and also matchhead compositions. Type II is intended for use in Tacot priming compositions. Classes 1, 2, 3, and 5 are intended for use in priming compositions. Class 4 is intended for use in matchhead compositions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.0).
c. Provisions for submission/resubmission of first article samples.
d. Quantity required and delivery schedules.
e. Type and class required (see 1.2).
f. Packaging, packing, and marking data with requirements in detail (see 5.1).
g. Item hazard classification.
h. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) tested.

6.3 Military unique. Antimony sulfide covered by this specification is a technical grade antimony sulfide used in military applications such as pyrotechnics, explosives, and primer. The particle size, moisture content, and other chemical characteristics required herein are significant.

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6.4 Material safety data sheets. Contracting officers should identify those activities requiring copies of the completed safety data sheets (SDS) prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313, and 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must identify the activities requiring copies of the SDS.

6.5 Definitions.

6.5.1 Batch. A batch is defined as that quantity of material that has been manufactured by some unit chemical process or subjected to some physical mixing intended to make the final product substantially uniform.

6.5.2 Gangue. The commercially valueless material in which ore is found.

6.6 Additional particle size requirements and analysis. Programs and suppliers may consider it desirable to impose additional requirements on particle size above and beyond those within the tolerances specified so as to specify further refinement of requirements or receive product optimized for their manufacturing processes (see 6.8). Furthermore, they may consider using alternate or additional methods and equipment to specify, characterize, and analyze size. Such analysis techniques may include:

a. Gravitational sedimentation (see ASTM B761 and ISO 10076)
b. Microscopy (see ISO 13322 and BS 3406-4)
c. Laser diffraction (see ASTM B882 and ISO 13320)

Those who are interested in alternate characterization techniques are cautioned that analysis and results from one technique are generally not directly comparable to other techniques due to numerous factors including particle morphology, measurement technique, etc. Beneficial information may be found on alternate techniques from the National Institute of Standards and Technology (NIST) Special Publication 960-1, Recommended Practice Guide on Particle Size Characterization, available at www.nist.gov.

6.7 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.8 Approval of equivalent test methods, instruments and/or reagents. Prior approval of the Contracting Officer should be required for use of equivalent test methods, instruments and/or reagents. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

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6.9 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.4) responsible for preparation of this specification.

6.10 Government limited liability. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.11 Subject term (key word) listing.

Explosives
Incendiaries
Metal
Primer
Propellants
Pyrotechnics

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army – AR

Preparing Activity:
Army - AR
(Project 6810-2021-010)

Review Activity:
Army - EA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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